

Controlling Diversity in Single-shot Motion Synthesis

Eleni Tselepi*

Electrical and Computer Engineering, University of
Thessaly
Greece
Moverse
Greece
etsselepi@e-ce.uth.gr

Georgios Albanis

Moverse
Greece
giorgos@moverse.ai

Spyridon Thermos

Moverse
Greece
spiros@moverse.ai

Anargyros Chatzitofis

Moverse
Greece
argyris@moverse.ai

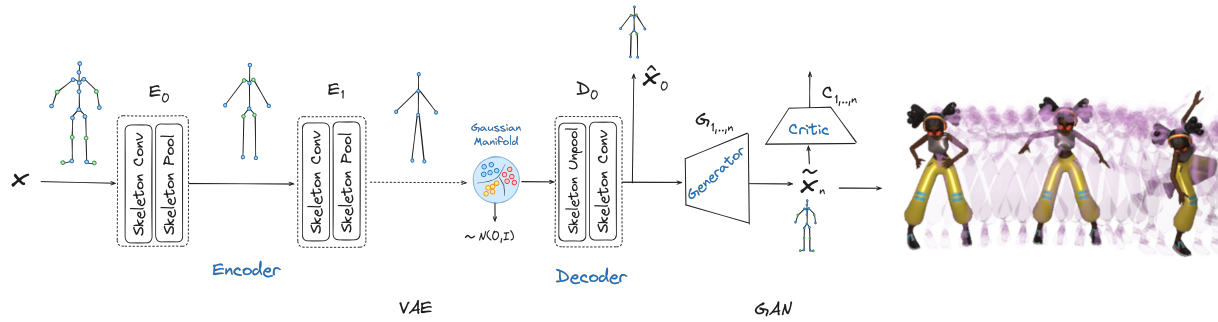


Figure 1: The proposed VAE-GAN for single-shot motion synthesis; Motion embeddings are folded around a multi-dimensional Gaussian manifold to learn a prior distribution of micro-motions, enabling controllable generation of motion variations.

Abstract

We consider the task of controllable and diverse motion synthesis from a single sequence as an alternative to the data-dependent text-to-motion methods, which pose ambiguities in data ownership and privacy. Recent works in hierarchical single-shot synthesis have paved the path for unconditional generation and editing tools, however the methods that focus on 3D animation have failed to control the diversity of the generated motions. In this paper we propose the integration of the variational inference in single-shot GANs, aiming to encode and control the low-frequency generating factors of the single motion sample. Our experiment showcases the ability of our VAE-GAN model to control the diversity of its generations, while preserving their plausibility and quality.

Keywords

Motion synthesis, 3D animation, diverse, single-shot, GAN

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1 Introduction

Generative models that create realistic variations of a single sample are powerful tools for editing and rapid prototyping, offering an alternative to the general-purpose text-to-modality approach. In fact, they have been successfully applied to a broad spectrum of applications, ranging from texture editing [Shocher et al. 2019] to motion synthesis [Li et al. 2022; Raab et al. 2024]. Specifically for motion synthesis, two predominant works have approached the problem using different backbone generative models: [Li et al. 2022] propose a Generative Adversarial Network (GAN) termed GANimator; [Raab et al. 2024] present a diffusion model with a U-Net backbone. Both works process a variety of features (per frame) such as rotation, foot joint velocity, root position, and concatenate them forming a generic motion feature representation which is used as the so-called “real” representation in the generative process. However, even though both models learn to generate high-quality variations of the targeted motion, none of them can control the diversity of the generation, reducing the actual impact of single-shot synthesis in applications such as 3D animation.

This work aims to circumvent the lack of controllability in single-shot motion generation by learning to decompose and map the input

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motion’s factors of variation on a Gaussian-shaped manifold. To achieve that, we follow [Gur et al. 2020] and propose a VAE-GAN architecture for single-shot motion synthesis.

2 VAE-GAN Architecture

We begin by describing the formation of the input motion feature representation and then present the proposed model architecture as depicted in Fig. 1. The overall model is a hierarchical VAE-GAN, with a variant of the traditional VAE focused on decomposing the factors of variation of a single input motion, and a GAN with a reduced receptive field adopted from [Li et al. 2022].

We adopt the **feature representation** from [Li et al. 2022] and form a motion feature $x_T \equiv \mathbb{R}^{T \times (JQ+C+3)}$, where J and Q are the number of skeleton joints and their 6D rotation representation, C corresponds to the foot contact labels that are followed by the root joint x - and z -axis velocity and the y -axis position. T corresponds to the temporal dimension, while for the rest of the paper, subscripts $\{i = 0, \dots, n\}$ correspond to the stage of the module/feature (*i.e.*, its temporal dimension level). **VAE module:** The traditional VAE ($f : E \circ D$) aims to minimize the ELBO criterion in order to balance the reconstruction error and the approximate posterior $q(x|z)$ from the Gaussian prior $p(z)$ by tuning the β hyperparameter (see Eq. 1). Our encoder E and decoder D consist of skeleton-aware convolution and pooling layers [Aberman et al. 2020], which encode input motion features into a reduced common representation, while also downsampling the temporal dimension. Note that since the following GAN operates in a coarse-to-fine motion, D upsamples z only once. For the **GAN module** ($f : G_\psi \circ C_\gamma$) we adopt the architecture of [Li et al. 2022], which learns to generate motion in a staged fashion (4 stages) using generators and discriminators - or critics- $\{G, C\}$ pairs. Starting from the output of the VAE module, the GAN learns to generate the motion features of increasing temporal resolution at each stage. Note that similar to the VAE module, G and C also use skeleton-aware convolutions, upsampling the intermediate motion features using skeleton-aware unpooling. Additionally, $\{E, D, G, C\}$ use a fixed receptive field $r = 1$. We train our single-shot VAE-GAN model using **loss functions** for: a) mapping the input motion embeddings to an “explorable” manifold, b) reconstructing the input motion, and c) preserving an equilibrium in the adversarial game between G and C . For all loss functions, the common constraint is the single motion input. Starting with the VAE, the first loss is the combination of the Kullback-Leibler (KL) divergence of the embedding and the Gaussian prior, and the reconstruction of the (downsampled) input:

$$\mathcal{L}_{vae} = \mathbb{E}_{q_\phi(z|x)} [\log(p(\theta|z))] - \beta \mathcal{D}_{KL}[q_\theta(z|x)||p(z)], \quad (1)$$

The $\{G, C\}$ are supervised using the Wasserstein variant from [Gulrajani et al. 2017]:

$$\mathcal{L}_{gan} = \min_{G_i} \max_{C_i} \mathbb{E}[C_i(\hat{x}_i) - C_i(x_i)] + \lambda_{gp} \mathbb{E}\left[\left(\|\nabla C_i(\tilde{x}_i)\|_2 - 1\right)^2\right],$$

where $\tilde{x}_i = \alpha \hat{x}_i + (1 - \alpha)x_i$ corresponds to the linear combination of the generated and the real motion features. To prevent mode collapse due to the single motion constraint, we optimize G with an $L1$ reconstruction loss, $\mathcal{L}_{rec} = \|G_i(\hat{x}_{i-1}, z_i^*) - x_i\|_1$, where z_i^* is a predefined noise embedding that is assigned to learn the reconstruction of x . Additionally, we opt to supervise the estimated

Table 1: Our random sampling and manifold walk vs. SoTA.

	SiFID (↓)	Global Div. (↑)	Local Div. (↑)	Inter Div. (↑)	Intra Div. (↑)
GANimator [Li et al. 2022]	1.14	1.02	0.97	1.47	2.13
SinMDM [Raab et al. 2024]	1.08	1.42	1.01	1.48	1.94
Ours	1.13	1.17	0.97	1.51	2.08
Ours + walk	1.12	0.82	0.97	1.15	2.07

foot-level joints’ contact C with the ground:

$$\mathcal{L}_{con} = \frac{1}{T|C|} \sum_{j \in \{C\}} \sum_{t=1}^T \|\mathcal{V}^{t,j}\|_2^2 \cdot S(C^{t,j}), \quad (2)$$

where S represents a skewed Sigmoid function to estimate the contact probability and $\mathcal{V}$ is the contact joint velocity. Our overall optimization approach can be summarized as:

$$\mathcal{L} = \lambda_{vae} \mathcal{L}_{vae} + \lambda_{gan} \mathcal{L}_{gan} + \lambda_{rec} \mathcal{L}_{rec} + \lambda_{con} \mathcal{L}_{con}, \quad (3)$$

where λ_* is a hyperparameter that tunes the contribution of the corresponding loss to the overall learning setup.

3 Experiments & Conclusions

We evaluate our VAE-GAN model against the state-of-the-art single-shot motion synthesis methods, namely the GANimator [Li et al. 2022] and the SinMDM ones [Raab et al. 2024], adopting their common benchmark from the available sequences from Adobe Mixamo. Table 1 presents the results of the following experiment; First we sample random codes from our Gaussian manifold and generate motion variants from Mixamo input samples, comparing our results in global and local diversity, as well as motion quality measured using a variation of the Fréchet Inception Distance (SiFID) but for a single data point. Note that in order to measure diversity, we adopt the respective metrics from both [Li et al. 2022] and [Raab et al. 2024]. We use this step, whose results are presented in rows 2-4 of Table 1, as the baseline for the second part of the experiment. In the second part, we sample only one code from the Gaussian manifold and generate the same amount of motion variants by adding or subtracting a constant value $\epsilon = 0.01$ to/from the sampled code. The success of our experiment lies in maintaining quality and local diversity during “manifold walking”, while reducing global diversity when sampling from the same manifold neighborhood. In fact, the last row of Table 1 shows that both global diversity metrics (depicted in red) decrease, while the quality of the generations and the local diversity of each generation are preserved.

Although our results are promising, they only begin to explore diversity control in single-shot motion synthesis. Future work will examine the manifold’s properties, its application impact, and expand evaluation with additional datasets and a potential user study.

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